System Design Specification: Modular Base CSE and Multi-Instance Vehicle AE Architecture

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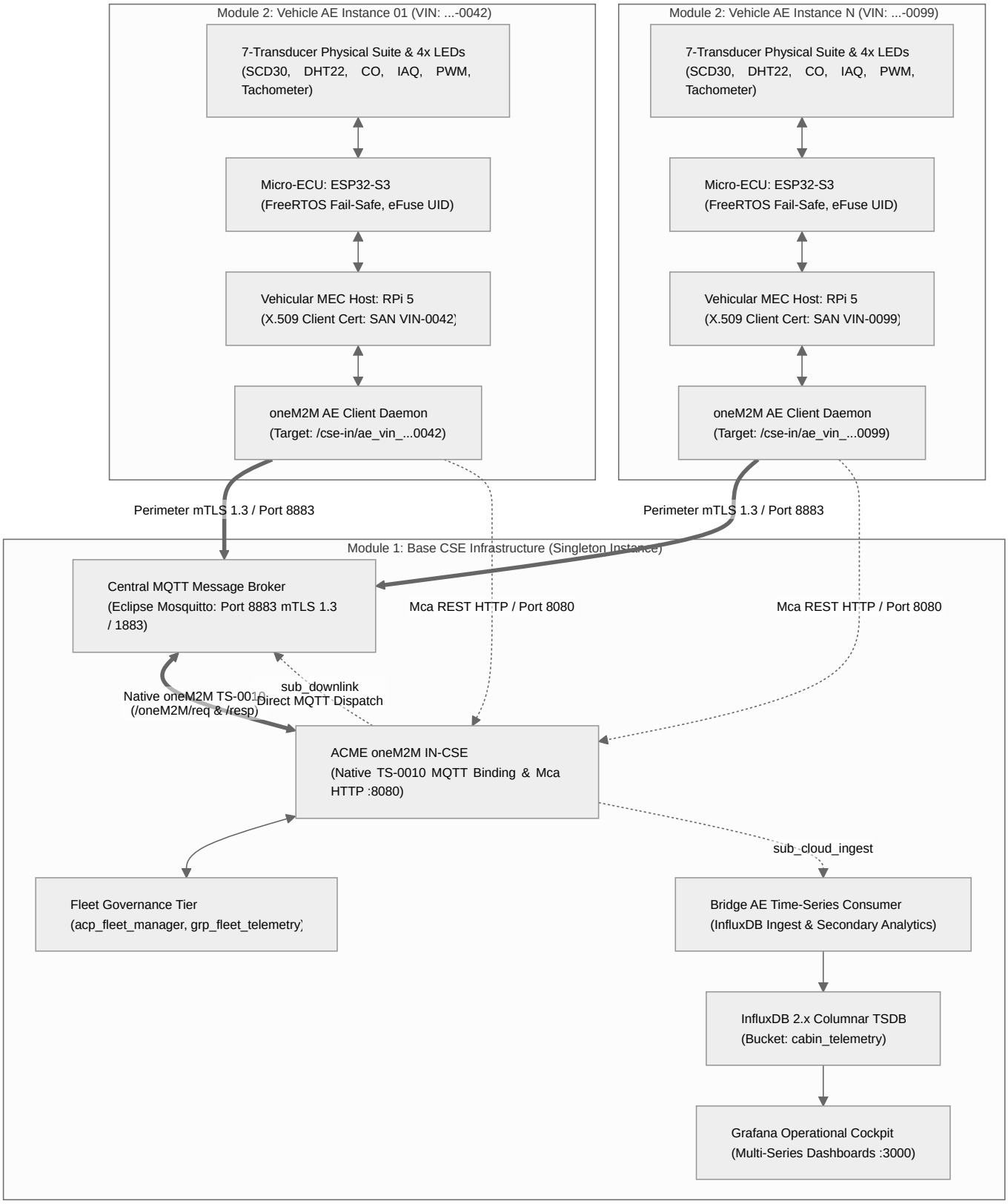
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Associated Baseline: acme_cse_gateway_capability_review.md

1. EXECUTIVE SUMMARY AND ARCHITECTURAL PRINCIPLES

This specification defines the structural partitioning, interface contracts, identity-binding protocols, and deployment lifecycle for the SDV Cyber-Physical Environmental Monitoring and Closed-Loop HVAC Regulation platform.

The system is decoupled into two strictly demarcated modules: 1. **Base CSE Infrastructure Module (Singleton Instance):** A centralised Common Services Entity (ACME oneM2M IN-CSE) serving as the authoritative cloud/edge registry, semantic middleware repository, fleet governance policy engine, and aggregated time-series telemetry ingest point. **This module is strictly a single, authoritative instance per operational cluster.** 2. **Vehicle Application Entity (AE) Module (Horizontally Scalable Multi-Instance):** An autonomous cyber-physical edge computing stack deployed onboard each physical vehicle. Each instance comprises physical transducers (SCD30, DHT22, CO, IAQ/VOC, AQI) interfaced via an ESP32-S3 microcontroller to an onboard vehicular MEC host (Raspberry Pi 5). **This module is deployable across N uniquely identified, concurrent, and independent instances.**



2. CONSOLIDATION OF VEHICULAR EDGE GATEWAY FUNCTIONS INTO ACME ONEM2M CSE

As established in the architectural capability review (acme_cse_gateway_capability_review.md), the platform consolidates legacy edge gateway microservices directly into native ETSI TS 118 101 middleware capabilities.

2.1 Excised Microservices and Absorbed Capabilities

1. **Decommissioning of Node-RED Ingress Proxy (`nodered_ipe`):** * *Legacy Role:* Acted as an intermediate Node.js container translating raw MQTT topics into HTTP REST `<m2m:cin>` instances, and translating HTTP webhooks into downlink MQTT actuation commands. * *Consolidated Architecture:* Completely decommissioned and excised from the deployment mesh. ACME oneM2M IN-CSE natively terminates and dispatches oneM2M primitives over MQTT via **oneM2M TS-0010 MQTT protocol bindings** (`[mqtt]` in `deploy/onem2m/acme.ini`). The vehicle node publishes ContentInstances directly to `/oneM2M/req/...` and receives actuation

downlinks on `/oneM2M/req/id-in/{aei}/json`. * *Performance Gain:* Reclaims ~220 MB RAM and reduces closed-loop actuation propagation latency by $40\text{--}80\ \text{ms}$.

2. **In-CSE Multi-Hazard Health Classification:** * *Legacy Role:* The standalone Python `bridge_ae` microservice evaluated all sensor divergence, stale timestamps, and motor stalls. * *Consolidated Architecture:* ACME CSE's internal scripting engine (`acme-scripts` with `onCreate` hooks) absorbs multi-hazard rule evaluation directly upon ContentInstance creation. The CSE tags `<cin>` resources with semantic labels (`lbl: ["health:FRESH", "delta_t:0.32", "delta_rh:1.8", "stall:false"]`) and raises emergency alarms internally. `bridge_ae` is streamlined into a lightweight, non-blocking time-series persistence consumer.
3. **Edge Time-Series Buffering and Storage Caps:** * *Consolidated Architecture:* Rather than running a heavyweight edge TSDB for short-term buffering, ACME CSE enforces standard oneM2M container lifecycle bounds:
   - `mni: 1440`: Buffers 1,440 historical instances (24 hours at 1-minute cadence).
   - `mbs: 2097152`: Strict 2 MB memory-footprint cap protecting vehicular storage.
   - `mia: 86400`: 24-hour automatic TTL instance pruning.
   - `missingDataDetect: true`: Built-in sensor loss detection without custom watchdog daemons.
4. **Dynamic Fleet Group Fan-Out (`<grp>`, `<fopt>`):** * *Consolidated Architecture:* Centralised group aggregation allows atomic fleet status queries (`/grp_fleet_telemetry/fopt/la`) and multi-vehicle broadcast actuations without external orchestrator scripts.
5. **Dynamic Semantic Policy Engine (`<acp>`):** * *Consolidated Architecture:* Replaces static broker ACL text files with standard oneM2M Access Control Policies, dynamically binding privileges to verified AE-IDs.

2.2 Architectural Segregation: Retained Dedicated Platforms

To guarantee cyber-physical safety, cryptographic robustness, and high-performance analytics, specific functions are strictly retained on dedicated platforms outside the CSE:

Subsystem	Placement	Architectural Rationale
Mutual TLS 1.3 Termination	**Eclipse Mosquitto (Port 8883)**	Cryptographic perimeter security, hardware ECDSA P-256 acceleration, and X.509 client certificate verification must terminate at a hardened C broker, protecting the Python ASGI/WSGI stack.
Long-Term Multi-Series TSDB	**InfluxDB 2.x Columnar Engine**	High-cardinality nanosecond time-series analytics and Flux queries across months of fleet operation exceed the capacity of oneM2M SQLite/TinyDB metadata backends.
Operational Cockpit	**Grafana Enterprise (Port 3000)**	Fleet-wide visual observability, multi-series dashboards, dynamic alerting, and operator control cockpits.
Autonomous Safety Overrides	**ESP32-S3 FreeRTOS Firmware**	Safety-critical fail-safe loops (e.g. 50% baseline purge ventilation during communication loss or CO emergency) execute deterministically in microcontroller firmware independent of edge host or network state.

3. MODULE 1: BASE CSE INFRASTRUCTURE SPECIFICATION (SINGLETON)

3.1 Scope and Singleton Constraint

The Base CSE module acts as the single source of semantic truth for the entire vehicular fleet. It must be deployed exactly once per environment. Deploying multiple Base CSE instances within the same operational domain is prohibited unless configured under explicit oneM2M Mcc Inter-CSE federation topologies.

3.2 Core Components

1. **ACME oneM2M IN-CSE (`sdv_onem2m_cse`):** * Exposes standard Mca HTTP(S) reference point on port `8080` and native oneM2M TS-0010 MQTT binding on internal port `1883`. * Manages the root `<m2m:cb>` (CSEBase) identified by `cse_id` (`id-in`) and accessible at path `/{{ cse_rn }}` (`/cse-in` or `/sdv-cse`). * Enforces semantic schema validation, resource lifecycles, and access control policies.
2. **Fleet Governance Layer (Pre-Provisioned Hierarchy):** * **Fleet Access Control Policy (`acp_fleet_manager`):** Establishes default administrative and onboarding privileges. * **Fleet Fan-Out Group (`grp_fleet_telemetry`):** An aggregate `<m2m:grp>` whose member IDs (`mid`) dynamically index the raw telemetry containers of all onboarded vehicles. Supports atomic fleet queries via `/grp_fleet_telemetry/fopt/la`. * **Downlink & Cloud Ingestion Subscriptions:** Dispatches asynchronous change events to subscriber targets.
3. **Data Health & Storage Pipeline:** * **Central Mosquitto Broker (`sdv_mosquitto_broker`):** Terminates network mTLS 1.3 on port 8883. * **Bridge AE Time-Series Consumer (`sdv_bridge_ae`):** Consumes `<subscription>` notifications and writes clean records to InfluxDB. * **InfluxDB 2.x TSDB (`sdv_influxdb`):** Historical time-series persistence. * **Grafana Operational Cockpit (`sdv_grafana`):** Observability and fleet-wide KPI visualisation.

3.3 Initial State Guarantee

Upon initial bootstrap, the Base CSE module **must contain zero vehicular Application Entities**. No vehicle node shall be present until dynamically onboarded by an authentic Vehicle AE module.

4. MODULE 2: VEHICLE AE SPECIFICATION (MULTI-INSTANCE)

4.1 Scope and Multi-Instance Constraint

The Vehicle AE module represents the onboard cyber-physical system of a single vehicle. Any arbitrary number of instances ($i \in \{1, 2, \ldots, N\}$) may be commissioned and deployed concurrently. Each instance is completely isolated from all other instances and maintains strict local autonomy.

4.2 Physical and Edge Compute Stack

Each Vehicle AE instance comprises two distinct computational tiers:

Tier A: Physical Sensing & Actuation Node (ESP32-S3 Micro-ECU)

- **Transducers (7 Environmental Channels):**
- Sensirion SCD30: NDIR CO_2 (0–10,000 ppm), dry-bulb temperature, relative humidity over I^2C Fast-Mode (400 kHz) with hardware CRC-8 verification.
- DHT22: Redundant temperature and relative humidity over 1-Wire bus with parity checksum.
- Carbon Monoxide (CO) Sensor: Toxic gas detection (0–1,000 ppm) with emergency alert interrupt.
- Cabin Air Quality (IAQ/VOC): Volatile organic compound index (0–500) for passenger air quality.
- Exterior Air Quality (AQI): Ambient outdoor index for fresh-air intake vs cabin air recirculation damper control (ΔIAQ).
- **Actuators (4-Channel Emulation Matrix & Closed-Loop Tachometer):**
- Blue LED (GPIO 18): Purge blower ventilation driven at 25.0 kHz PWM with 10-bit resolution ($0\text{--}100\%$ duty cycle).
- Red LED (GPIO 17): Auxiliary cabin PTC heating core driven by digital output ($T_{\text{cabin}} < 19.0\,^{\circ}\text{C}$).
- Green LED (GPIO 16): Baseline eco-ventilation driven by digital output (nominal air quality).
- Amber LED (GPIO 15): Dehumidification and warning alert driven by digital output ($RH > 65\%$ or degraded data health).
- WS2812B RGB LED (GPIO 38): Optical state annunciator indicating network, sensor, and safety status.
- Hall-Effect Tachometer Feedback (GPIO 19): Interfaced via Pulse Counter (PCNT) Unit 0 for closed-loop blower stall detection.
- **Firmware Runtime:** FreeRTOS with strict core affinity (Core 1: Sensor/PWM loop; Core 0: mTLS/Network loop).
- **Hardware Identity:** Immutable 48-bit MAC address and 128-bit UID read from write-once eFuse registers.

Tier B: Vehicular MEC Host (Raspberry Pi 5)

- **Compute Platform:** Quad-core ARM Cortex-A76 vehicular edge host.
- **Local Broker / Ingress Agent:** Manages local sensor aggregation, payload formatting, and schema validation.
- **Vehicle Identity Master:** Stores the vehicle's ISO 3779 VIN, OTP board serial, and cryptographic private keys.
- **oneM2M AE Client Agent:** Executes the onboarding reconciliation loop, periodic telemetry transmission to the Base CSE via native TS-0010 MQTT or Mca REST, and actuation downlink execution.

4.3 Dynamic oneM2M Resource Representation

When an instance registers with the Base CSE, it provisions a dedicated subtree under `<m2m:cb>`:

```
/cse-in (CSEBase)
 └── ae_vin_wauzzz_sdv_2026_0042 (Application Entity: <m2m:ae>)
      ├── temperature-cabin (Container: <m2m:cnt>)
      ├── humidity-cabin (Container: <m2m:cnt>)
      ├── co2-cabin (Container: <m2m:cnt>)
      ├── co-cabin (Container: <m2m:cnt>)
      ├── air-quality-cabin (Container: <m2m:cnt>)
      ├── air-quality-exterior (Container: <m2m:cnt>)
      ├── raw-telemetry (Container: <m2m:cnt>)
      │    ├── cin_xxxx (<m2m:cin>)
      │    └── sub_cloud_ingest (<m2m:sub>)
      └── commands (Container: <m2m:cnt>)
           └── sub_downlink_actuation (<m2m:sub>)
```

5. MULTI-LAYER BINDING MECHANISM: UNIQUELY LINKING VEHICLE TO AE

To guarantee that each physical vehicle solely and accurately links to its designated AE instance on the Base CSE without cross-talk or collision, the 5-layer binding mechanism operates across the module boundary:

Layer	Binding Mechanism	Implementation in Module Partition
1. Physical Hardware	ESP32 eFuse MAC + RPi 5 Board Serial $\rightarrow$ ISO 3779 VIN	**Vehicle AE Module:** Embedded in local device tree and sensor frame headers.
2. Transport Security	X.509 Mutual TLS 1.3 with Subject Alternative Name (SAN)	**Boundary Interface:** Vehicle AE presents cert with `SAN: URI:urn:sdv:vin:<VIN>`. Base CSE / Gateway authenticates cert against vehicular Root CA.
3. Semantic Identity	Deterministic Resource Name ($rn = \text{FNV-1a}(\text{VIN})$)	**Boundary Interface:** Vehicle registers/reconciles deterministic rn. Base CSE binds allocated aei to client credential.
4. Access Control	Dedicated `<m2m:acp>` per vehicle ($acop = 63$ for aei)	**Base CSE Module:** Restricts CRUD operations on `/ae_vin_<slug>/*` strictly to that vehicle's authenticated aei and $CAdmin$.
5. Downlink Routing	Targeted Point of Access (poa) & MQTT topics	**Boundary Interface:** Actuator directives route strictly to `sdv/{vin}/control/cmd` or native topic `/oneM2M/req/id-in/{aei}/json`.

5.1 Deterministic Resource Name Equation

To prevent collisions and eliminate hardcoded aliases, the AE Resource Name is mathematically derived from the vehicle's unique identity string:

$$rn_{\text{AE}} = \text{ae_vin_} \parallel \text{slugify}\ (\text{VIN}) \parallel \text{_} \parallel \text{hex}\ \left(\quad FNV-1a_{32}(\text{VIN})\ \right)$$

This guarantees: 1. **Uniqueness:** Two distinct vehicles ($VIN_A \neq VIN_B$) yield distinct resource names ($rn_A \neq rn_B$). 2. **Idempotence:** A rebooting vehicle always resolves the exact same resource name, eliminating duplicate "ghost" AEs.

6. MULTI-DIMENSIONAL DATA HEALTH CLASSIFICATION ENGINE

The system evaluates incoming sensor telemetry against five quantitative health criteria:

$$\Delta T = \ \mid T_{\text{SCD30}} - T_{\text{DHT22}} \mid \ \leq 1.5\,^{\circ}\text{C}$$

$$\Delta RH = \ \mid RH_{\text{SCD30}} - RH_{\text{DHT22}} \mid \ \leq 10.0\,\%$$

$$\tau_{\text{age}} = t_{\text{eval}} - t_{\text{sample}} \leq 2.5\,\text{s}$$

$$\text{Stall Fault} \iff (\text{PWM} \geq 30\%) \land (\text{RPM} < 100)$$

$$\text{Toxic Gas Hazard} \iff \text{CO} \geq 30.0\,\text{ppm}$$

Categorical State Assignment

1. **`FRESH`**: $\tau_{\text{age}} \leq 2.5\,\text{s}$, $\Delta T \leq 1.5\,^{\circ}\text{C}$, $\Delta RH \leq 10\%$, and tachometer feedback confirmed. Full closed-loop regulation active.
2. **`STALE`**: $2.5\,\text{s} < \tau_{\text{age}} \leq 10.0\,\text{s}$. Network latency delay; hold-last-valid-value regulation logic engaged.
3. **`DEGRADED`**: $\Delta T > 1.5\,^{\circ}\text{C}$ or $\Delta RH > 10.0\%$. Transducer divergence; amber annunciator illuminated; system relies on conservative SCD30 readings.
4. **`FAULT`**: $\tau_{\text{age}} > 10.0\,\text{s}$, null reading, out-of-bounds reading, or blower motor stall detected. Numerical values excluded from time-series aggregates; blower defaults to 50% baseline fail-safe ventilation.

7. DEPLOYMENT AND OPERATIONAL SCENARIOS

7.1 Scenario A: Singleton Base CSE Bootstrap

1. The infrastructure deploys `deploy/docker-compose.yml`.
2. `sdv_onem2m_cse`, `sdv_mosquitto_broker`, `sdv_bridge_ae`, `sdv_influxdb`, and `sdv_grafana` initialize (Node-RED is decommissioned).
3. The Base CSE governance bootstrap script (`deploy_acme_fleet_cse_base.py`) executes: * Verifies CSEBase reachability (`GET /cse-in`). * Provisions `acp_fleet_manager` (). * Provisions `grp_fleet_telemetry` () with an empty member array (`mid = []`). * **Result:** Clean, production-ready Base CSE with zero registered vehicle AEs.

7.2 Scenario B: Commissioning a New Vehicle AE Instance (N-th Instance)

Each physical vehicle executes its onboarding sequence independently: 1. **Hardware Power-On:** ESP32 boots FreeRTOS, begins sensor sampling, and establishes local bus communication with the Raspberry Pi 5. 2. **Identity Resolution:** Onboard agent reads local VIN and computes deterministic rn_{AE}. 3. **Reconcile-on-Boot Check:** * Vehicle issues `GET /cse-in/{{ vehicle_ae_rn }}` with `X-M2M-`

`Origin: CAdmin` (or authenticated client cert). * **Branch 1 (New Vehicle - 404 Not Found):** * Issues `POST /cse-in` with `ty=2` to create `<m2m:ae>` using deterministic rn. * Iterates through the canonical sensor manifest to create the 8 standard containers (`<m2m:cnt>`). * Attaches downlink subscription to `/commands`. * Registers its `raw-telemetry` container in Base CSE's `grp_fleet_telemetry.mid`. * **Branch 2 (Reconnecting Vehicle - 200 OK):** * Validates existing container tree. * Reclaims its assigned `aei` and immediately resumes telemetry commits. 4. **Steady-State Ingestion:** Vehicle posts telemetry instances (`<m2m:cin>`) via native oneM2M TS-0010 MQTT or Mca REST directly to its own `/raw-telemetry` container.

7.3 Scenario C: Decommissioning / Retiring a Vehicle AE Instance

When a vehicle is decommissioned or retired from the fleet: 1. Administrative command or vehicle departure agent issues `DELETE /cse-in/{{ vehicle_ae_rn }}`. 2. Base CSE cascade-deletes the vehicle's `<m2m:ae>`, all 8 child `<m2m:cnt>` containers, all `<m2m:cin>` records, and subscriptions. 3. Governance reconciliation script removes the deleted container URI from `grp_fleet_telemetry.mid`. 4. Other active vehicle AE instances remain entirely unaffected.

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8. ENVIRONMENT AND CONFIGURATION CONTRACTS

8.1 Base CSE Environment (`deploy/.env`)

The Base CSE host maintains infrastructure-wide variables:

```
# Infrastructure & Gateway Addressing
GATEWAY_IP=192.168.122.91
ONEM2M_PORT=8080
ONEM2M_CSE_ID=id-in
ONEM2M_CSE_NAME=cse-in

# Security & Governance
BASE_CSE_ADMIN_ORIGINATOR=CAdmin
FLEET_ACP_NAME=acp_fleet_manager
FLEET_GROUP_NAME=grp_fleet_telemetry

# Service Endpoints
MOSQUITTO_PORT=8883
INFLUXDB_HOST_PORT=8087
GRAFANA_PORT=3000
```

8.2 Vehicle AE Node Environment (`vehicle/.env` per vehicle)

Each physical vehicle host maintains its local instance parameters:

```
# Vehicle Hardware & Cryptographic Identity
VEHICLE_VIN=VIN-WAUZZZ-SDV-2026-0042
VEHICLE_NODE_ID=sdv-car-0042
CLIENT_CERT_PATH=/etc/sdv/certs/client.crt
CLIENT_KEY_PATH=/etc/sdv/certs/client.key

# Target Base CSE Gateway Connection
BASE_URL=http://192.168.122.91:8080
CSE_RN=cse-in
ORIGINATOR=CAdmin
APP_ID=N.org.fhtw.sdv.cabin
RVI=3

# Local Actuation & Downlink Configuration
LOCAL_ESP32_PORT=/dev/serial/by-id/usb-Espressif_ESP32-S3_001
DOWNLINK_NOTIFICATION_URL=mqtt://192.168.122.91:8883/oneM2M/req/id-in/C_ae_vin_wauzzz_0042/json
```

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9. QUALITY ATTRIBUTES AND VERIFICATION MATRIX

Quality Attribute	Requirement Specification	Verification Method
Singleton Guarantee	Exactly one Base CSE exists per network domain. Zero duplicate CSEBase roots permitted.	Unit & integration test `make validate-base-cse` verifying singular `<m2m:cb>`.
AE Scalability	Fleet capacity scales to N independent vehicle AEs without code modifications.	Iterative test harness executing onboarding for M arbitrary VINs.
Collision Immunity	Two vehicles with different VINs must never claim or overwrite the same AE tree.	Ingestion collision test asserting HTTP 409 / 403 on duplicate registration attempts.
Zero Initial State	Fresh deployment of Base CSE must start with zero vehicular AEs.	Preflight audit confirming `len(cb.childResource[ty==2]) == 0`.
Native Ingestion	Zero intermediary proxy containers; native oneM2M TS-0010 MQTT / Mca HTTP.	Integration test verifying Node-RED absence and direct primitive routing.
Fail-Safe Containment	Network disconnection of Vehicle A has zero impact on Vehicle B's ingestion loop.	Network fault injection testing with `inject_faults.py`.